

Practical Demo: Callable-Bond Valuation, OAS, and Effective Risk

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This reading copy shows the calculations' results and figures. The HTML demo contains the full code in folded blocks and interactive scenario controls.

The desk question

How much of a callable bond's apparent yield advantage compensates the investor for the issuer's exercise right?

Learning Objectives

After completing this demo you should be able to:

- value option-free and callable bonds with backward induction;
- apply the issuer's call decision at each eligible node;
- isolate the embedded call option's price cost;
- solve for the option-adjusted spread (OAS) that matches a market price; and
- calculate effective duration and effective convexity by repricing the bond.

1 Market Inputs Versus Illustrative Inputs

The analyst's question: a callable bond offers more yield than a straight bond. How much is compensation for surrendering refinancing upside? Follow **cash flows** → **rate states** → **issuer exercise** → **fitted spread** → **risk**. At every node, think from the issuer's perspective: the investor does not get to choose the more valuable outcome.

This is a transparent teaching workflow, not a live security valuation. A production model would calibrate an arbitrage-free tree to a current benchmark curve and an option-volatility surface.

Input	Value	Status
Face value	100	Illustrative bond term
Annual coupon rate	5.25%	Illustrative bond term
Final maturity	7 years	Illustrative bond term
Call schedule	Years 3–6	Illustrative bond term
Call prices	101.50, 101.00, 100.50, 100.00	Illustrative bond term
Median one-year short-rate path	4.20% to 3.70%	Market-like, not live
Annual short-rate volatility	18%	Illustrative model input
Risk-neutral up probability	50%	Simplifying assumption
Observed callable-bond price	Generated below	Illustrative market price

The rate tree is recombining and lognormal. It captures path-dependent call decisions, but it is not presented as a fully calibrated term-structure model. Rates cannot be negative, the time step is one year, the OAS is a constant spread, and transaction costs, taxes, accrued interest, default, recovery, and notice periods are ignored.

Workflow stage	In this case
Supplied in the teaching case	Callable price and contractual cash flows
Assumed	Benchmark rate tree, volatility, and exercise convention
Solved	OAS that reprices the observed bond price
Diagnosed	Repricing error and call exercise by node
Used for	Effective duration, convexity, and rate-stress decisions

2 Build a Recombining Short-Rate Tree

At time t , node j has short rate

$$r_{t,j} = m_t \exp[(2j - t)\sigma],$$

where m_t is the median-rate path and σ controls annual volatility. Moving up increases j by one; moving down leaves j unchanged. Because the same number of up moves leads to the same node, the tree recombines.

	Year 0	Year 1	Year 2
Node (number of up moves)			
0	4.2	3.425	2.791
1		4.909	4
2			5.733
3			
4			
5			
6			

	Year 3	Year 4	Year 5
Node (number of up moves)			
0	2.273	1.85	1.525
1	3.258	2.651	2.185

	Year 3	Year 4	Year 5
Node (number of up moves)			
2	4.669	3.8	3.132
3	6.692	5.447	4.49
4		7.807	6.435
5			9.224
6			

	Year 6
Node (number of up moves)	
0	1.257
1	1.801
2	2.581
3	3.7
4	5.303
5	7.601
6	10.895

The entries are annual short rates in percent. The low-rate nodes are economically important because they are where refinancing and call exercise become most attractive.

3 Price the Option-Free and Callable Bonds

Values are measured immediately after any coupon at a node. The one-period holding value is

$$V_{t,j}^{\text{hold}} = \frac{C + pV_{t+1,j+1} + (1-p)V_{t+1,j}}{1 + r_{t,j} + s}.$$

For the option-free bond, this holding value is retained. At an eligible call date, the callable bond applies the issuer's minimum rule:

$$V_{t,j}^{\text{callable}} = \min \{V_{t,j}^{\text{hold}}, \text{Call Price}_t\}.$$

```
Option-free value    107.0091
Callable value       103.3419
Embedded call value   3.6672
dtype: float64
```

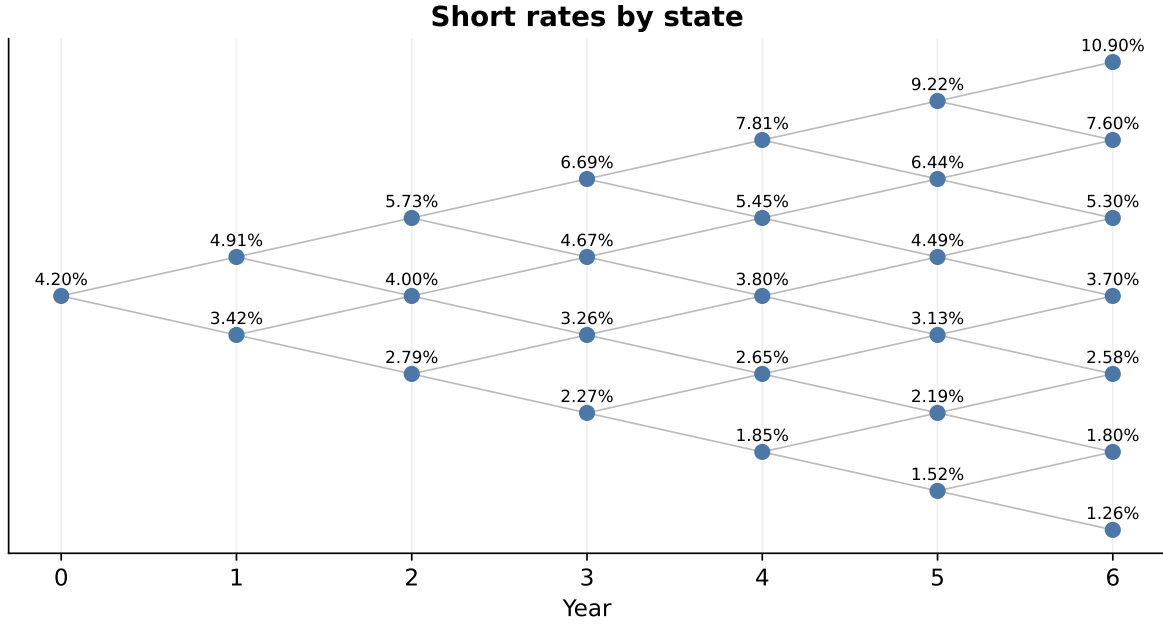


Figure 1: Illustrative recombining short-rate tree

The price decomposition is

$$V_{\text{callable}} = V_{\text{option-free}} - V_{\text{call option}}.$$

The option cost is positive because the issuer owns the call. It is largest when low-rate paths make refinancing valuable.

4 Illustrate and Solve for OAS

4.1 See where the issuer calls

At year 3, compare the ex-coupon continuation value with the 101.50 call price. If continuation is worth more than 101.50 to the investor, the issuer can retire that liability for 101.50 instead. The coupon due at that date has already been paid under our timing convention; do not add it to the call price again.

To make the exercise reproducible, suppose the callable bond trades at the model value generated by an 85-basis-point spread. We then hide that spread and recover it from the price. In practice, the price would come from the market.

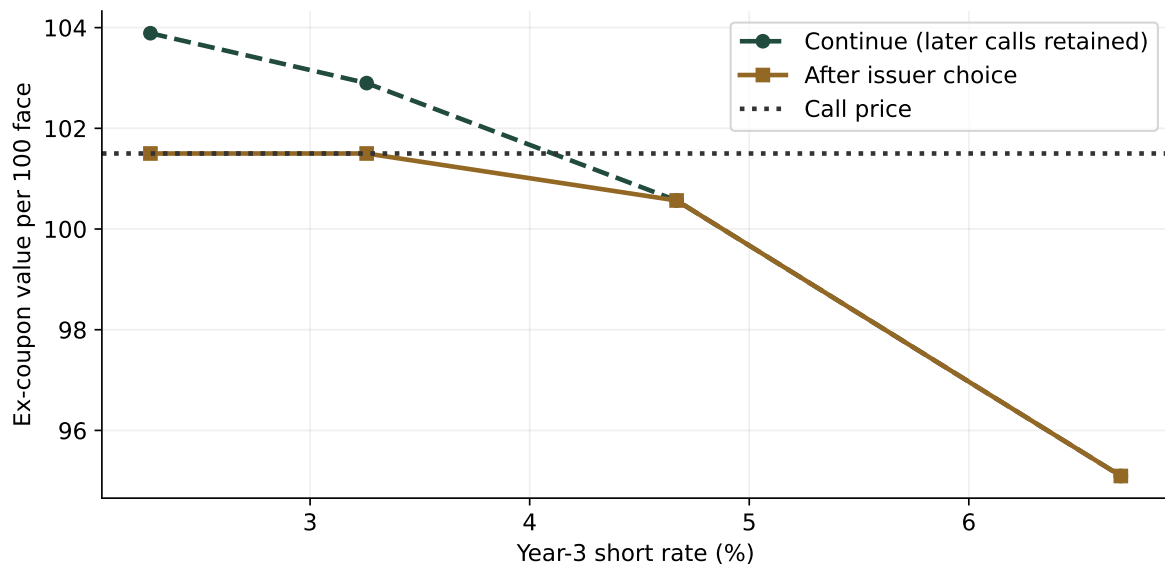


Figure 2: Year-3 exercise decision at zero spread: the investor receives the lesser of continuation and the call price.

Illustrative market price	100.0700
Solved OAS (bp)	85.0000
Static spread on promised cash flows (bp)	115.8285
Option cost in spread terms (bp)	30.8285
Repriced callable value	100.0700
dtype: float64	

OAS is the spread left after the tree has modeled the issuer's exercise decision. The static spread is larger because it discounts promised maturity cash flows and does not recognize that favorable low-rate paths can be taken away through a call.

The option cost can also be expressed in price terms at the solved OAS:

Option-free value at OAS	101.8603
Callable value at OAS	100.0700
Call option cost in price points	1.7903
dtype: float64	

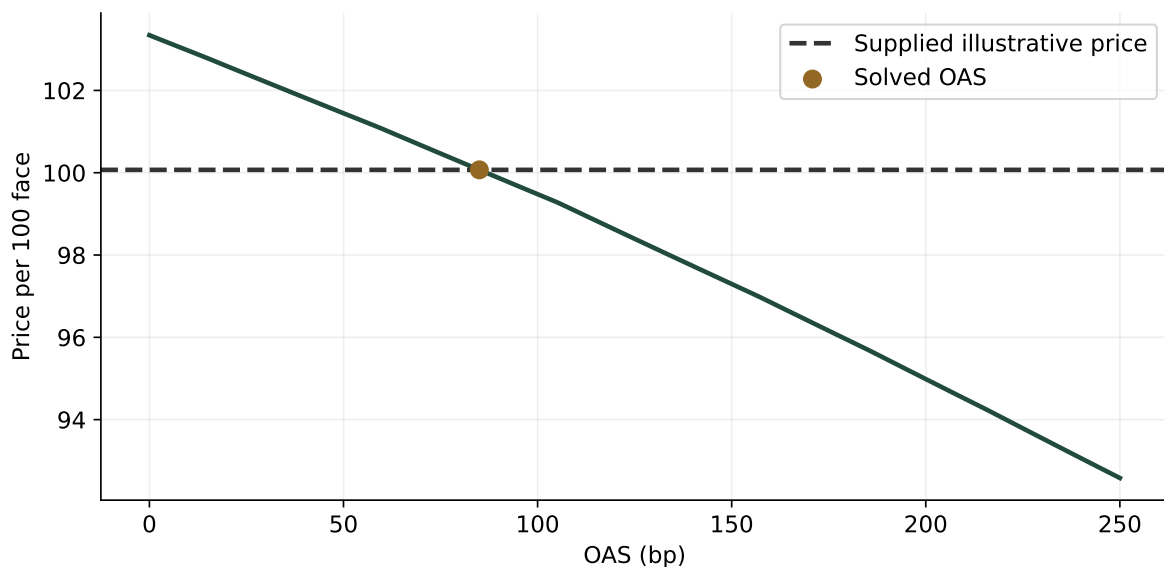


Figure 3: Solving OAS means moving along the callable-price curve until model and supplied price agree.

5 Effective Duration and Effective Convexity

💡 Predict before repricing

If the target price falls while the tree and contract stay fixed, should the solved spread rise or fall? Use the figure to explain the bisection direction. For a rate shock, hold OAS fixed: re-solving OAS to the old price would erase the very price sensitivity you are trying to measure.

Modified duration is not sufficient because the call decision changes across rate paths. Hold OAS fixed, shift the entire benchmark tree by $\pm\Delta y$, and fully reprice:

$$D_{\text{eff}} = \frac{P_- - P_+}{2P_0\Delta y},$$

$$C_{\text{eff}} = \frac{P_- + P_+ - 2P_0}{P_0(\Delta y)^2}.$$

	P0	P(-25 bp)	P(+25 bp)
Option-free bond	101.8603	103.3417	100.4054

	P0	P(-25 bp)	P(+25 bp)
Callable bond	100.07	101.0662	99.0611

	Effective duration	Effective convexity
Option-free bond	5.7655	41.7602
Callable bond	4.0074	-20.2338

The callable bond should generally have shorter effective duration because its upside is capped when rates fall. Its effective convexity can also be much lower, or negative, near the region where call exercise changes rapidly.

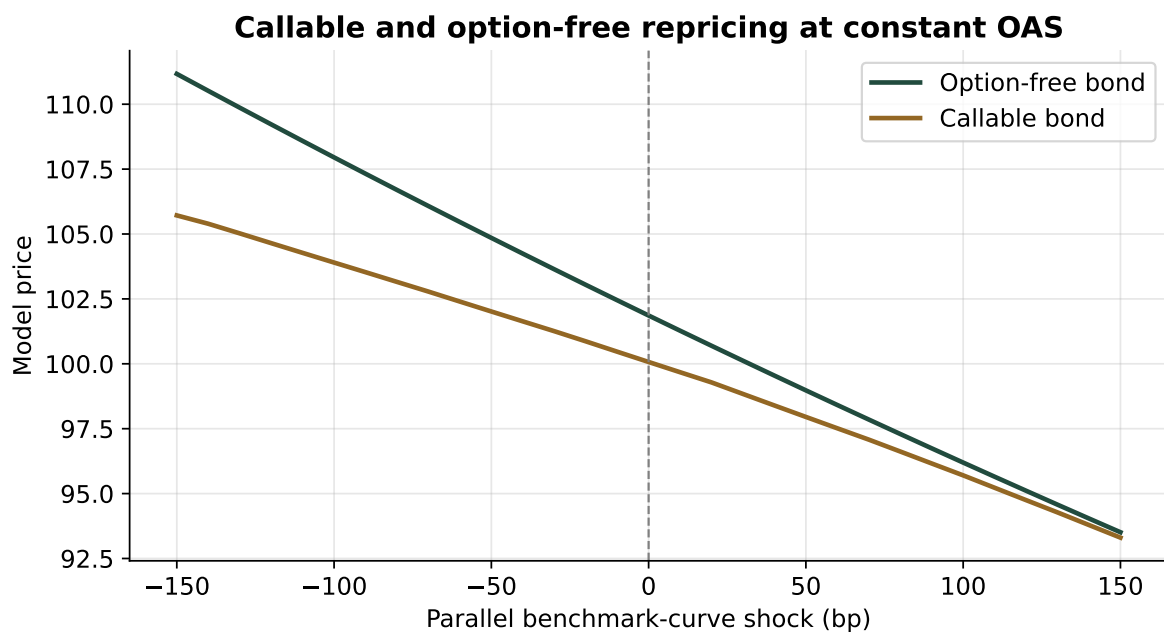


Figure 4: Full repricing reveals the callable bond's capped upside

6 Validation Checks

OAS recovered within 0.0001 bp	True
Model reprices market value	True
Callable value does not exceed straight value	True
Price falls when the curve rises	True

```
Price rises when the curve falls           True
dtype: bool
```

7 Financial Interpretation

This workflow separates three ideas that a yield-to-maturity comparison mixes together:

1. The benchmark short-rate tree values state-contingent cash flows.
2. The minimum rule transfers low-rate upside from the investor to the issuer.
3. OAS measures the constant spread that remains after the call option has been modeled.

The outputs are model-dependent. A different benchmark curve, volatility, call behavior, or tree calibration can materially change the option value, OAS, effective duration, and effective convexity.

7.1 Market handoff

Inspect a callable issue's terms and recent trades using [FINRA Fixed Income Data](#), then verify the complete call schedule and notice provisions in the offering document. Use a dirty price consistently with the valuation date. This notebook's generated price and median-rate tree are the offline teaching case, **not a calibration to an observed security**. The [curve-building demo](#) shows the separate benchmark-curve calibration step.

Your deliverable: price, OAS, call-option cost, effective duration, and a two-sentence explanation of which inputs must be refreshed before trading.

8 Interactive explorer

Move the benchmark tree while holding the recovered OAS fixed. Compare the callable and option-free bonds. In falling-rate states the issuer takes away part of the upside. This is a parallel shift of the teaching tree, not a recalibration of a market yield curve.

The HTML control selects a precomputed scenario. In the PDF, use the preceding figures and tables and rerun the displayed notebook inputs to explore other values.

9 Mini Project

Project brief. Using the same annual tree, compare option-free and callable values at spreads of 0, 50, and 100 bp. For \$3 million face, report each price, the embedded option cost per 100 and in dollars, and the price change from 0 to 100 bp. Explain why the two bonds must be valued at the same spread when isolating option cost. Distinguish this spread shock from a recalibrated benchmark-curve shock.

Deliverable: a one-page decision memo with a conventions statement, reproducible calculations, a scenario table, and a limitation that could change the decision.

Worked solution: calculations and decision memo

	Spread (bp)	Straight	Callable
0	0	107.0091	103.3419
1	50	103.9419	101.4455
2	100	100.9842	99.479

	Call cost per 100	Call cost on \$3mm
0	3.6672	110,015.9037
1	2.4964	74,892.6196
2	1.5052	45,156.8429

```

Straight 0-to-100 bp P&L ($)  -180748.23
Callable 0-to-100 bp P&L ($)  -115889.17
dtype: float64

```

Subtract callable value from straight value using identical cash-flow conventions, rates, and spread. Otherwise the difference includes a spread effect as well as optionality. The issuer's minimum rule limits the investor's upside. The spread exercise holds the tree fixed; a benchmark-curve shock requires rebuilding or recalibrating the tree and holding OAS fixed for effective risk. This illustrative tree does not fit a supplied market curve, so it supports mechanics rather than a production OAS estimate.